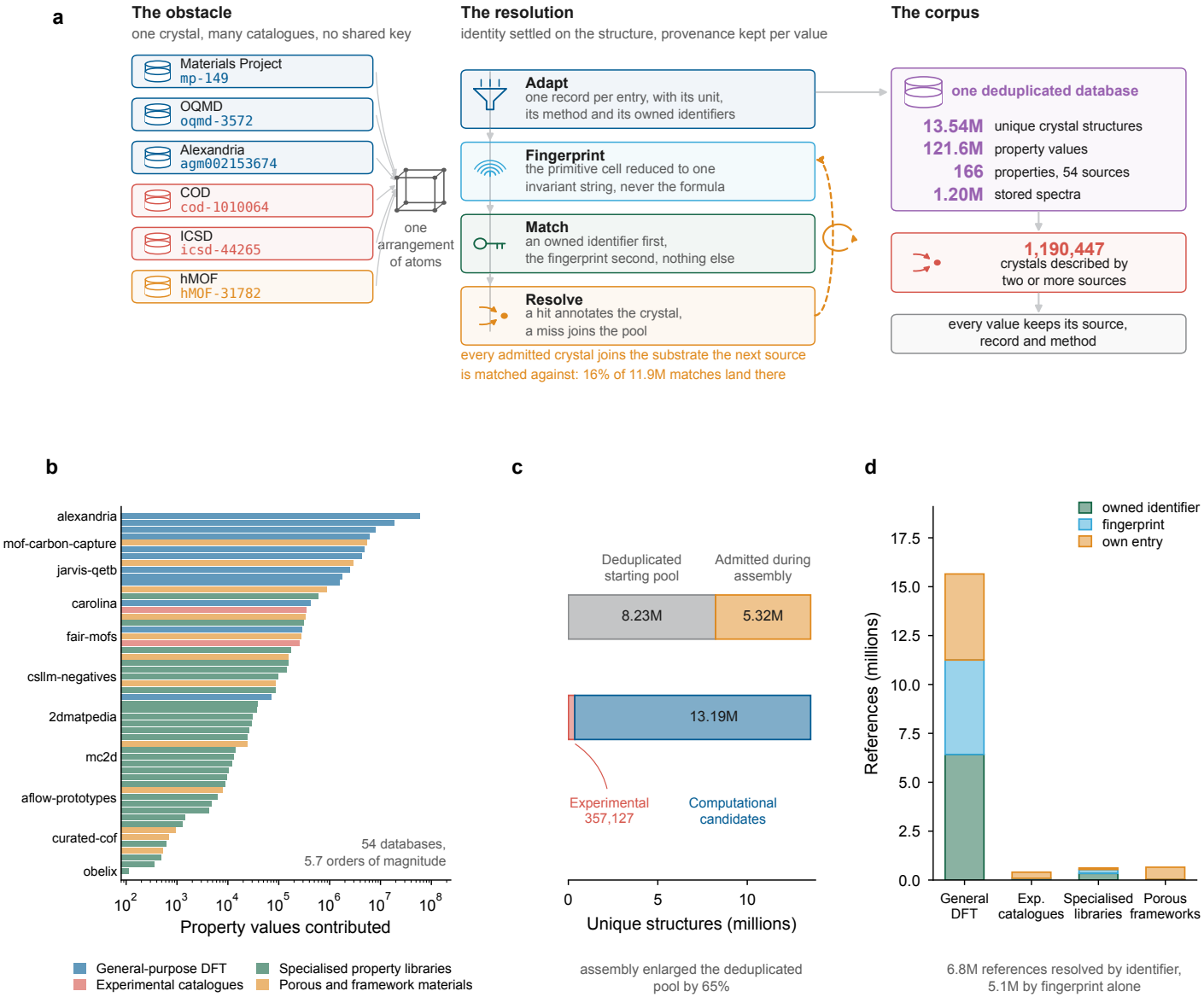

a
The obstacle
one crystal, many catalogues, no shared key
Materials Project
mp-149
OQMD
oqmd-3572
Alexandria
agm002153674
COD
cod-1010064
ICSD
icsd-44265
hMOF
hMOF-31782
one arrangement of atoms
The resolution
identity settled on the structure, provenance kept per value
Adapt
one record per entry, with its unit, its method and its owned identifiers
Fingerprint
the primitive cell reduced to one invariant string, never the formula
Match
an owned identifier first, the fingerprint second, nothing else
Resolve
a hit annotates the crystal, a miss joins the pool
every admitted crystal joins the substrate the next source is matched against: 16% of 11.9M matches land there
The corpus
one deduplicated database
13.54M unique crystal structures
121.6M property values
166 properties, 54 sources
1.20M stored spectra
1,190,447
crystals described by two or more sources
every value keeps its source, record and method
b
alexandria
mof-carbon-capture
jarvis-qetb
carolina
fair-mofs
csllm-negatives
2dmatpedia
mc2d
aflow-prototypes
curated-cof
obelix
54 databases,
5.7 orders of magnitude
Property values contributed
General-purpose DFT
Experimental catalogues
Specialised property libraries
Porous and framework materials
c
Deduplicated starting pool
Admitted during assembly
8.23M
5.32M
13.19M
Experimental
357,127
Computational candidates
Unique structures (millions)
assembly enlarged the deduplicated pool by 65%
d
owned identifier
fingerprint
own entry
References (millions)
General DFT
Exp. catalogues
Specialised libraries
Porous frameworks
6.8M references resolved by identifier, 5.1M by fingerprint alone